

## Chapter 9: Hyper-Parameter Tuning & Synthetic Data Analysis

Advances in Financial Machine Learning (Ch. 9) & Book 2 (Appendices A & B)

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## Chapter 9 Overview: The Tuning Challenge in Finance

### The Need for Hyper-Parameter Tuning

- **Model Generalization:** Hyper-parameters (e.g.,  $C$  and  $\gamma$  in SVMs, tree depth) control capacity and regularization.
- **The Financial Trap:** Tuning hyper-parameters using standard  $k$ -fold CV causes severe **information leakage** across overlapping time labels.
- **Sub-optimal Scoring:** Using standard Accuracy as a tuning metric rewards unconfident hits and ignores catastrophic high-confidence misses.

### Key Solutions in Chapter 9

- **Purged  $k$ -Fold Integration:** Combining GridSearchCV and RandomizedSearchCV with PurgedKFold (purging + embargoing).
- **Log-Uniform Random Search:** Efficiently searching vast, multi-dimensional hyper-parameter spaces under a computational budget.
- **Log-Loss (neg\_log\_loss):** Scoring models by probability-weighted accuracy to align tuning with PnL risk.

## Topic 9.1: Grid Search with Purged K-Fold Cross-Validation

### Preventing Leakage During Hyper-Parameter Optimization

- **Standard GridSearchCV Failure:** Evaluates exhaustive parameter combinations using standard  $k$ -fold CV, leaking future information through overlapping labels and serial correlation.
- **The Purged Solution:** Pass `PurgedKFold(n_splits=cv, t1=t1, pctEmbargo=pctEmbargo)` into `GridSearchCV`.

### Scoring Metric Selection Rule

$$\text{Scoring Metric} = \begin{cases} \text{'f1'} & \text{for Meta-Labeling applications (classifying } y_i \in \{0, 1\}) \\ \text{'neg\_log\_loss'} & \text{for General directional/sizing applications} \end{cases}$$

- Ensures that hyper-parameter choices optimize out-of-sample probability calibration rather than in-sample noise.

## Topic 9.2: Randomized Search Cross-Validation

### Why Grid Search Fails in High Dimensions

- **Grid Explosion:** If an algorithm has  $D$  hyper-parameters with  $M$  search values, grid search requires  $M^D$  model evaluations.
- **Irrelevant Parameters:** Grid search wastes computational time testing granular steps for parameters that have minimal performance impact.

### The Randomized Search Advantage

- **Fixed Computational Budget:** User sets `n_iter`, decoupling search time from parameter dimension  $D$ .
- **Log-Uniform Distribution:** Sampling continuous parameters (e.g., learning rates, regularization  $C$ ) uniformly on a logarithmic scale:

$$X \sim \text{LogUniform}(a, b) \implies \ln(X) \sim U(\ln a, \ln b)$$

Ensures equal representation across order-of-magnitude scales  
( $10^{-2}$ ,  $10^{-1}$ ,  $1$ ,  $10^1$ ,  $10^2$ ).

## Topic 9.3: Scoring Metrics – Why Log-Loss Beats Accuracy

### The Fallacy of Accuracy in Financial ML

- Accuracy treats a wrong prediction with 51% confidence the same as a wrong prediction with 99% confidence.
- In trading, positions are sized by confidence ( $m \propto p$ ). A high-confidence miss results in catastrophic drawdown, whereas Accuracy ignores probability calibration.

### Log-Loss / Cross-Entropy Loss Formulation

$$L[Y, P] = -\frac{1}{N} \sum_{i=1}^N [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)]$$

- Exponentially penalizes confident errors (e.g., predicting  $p_i = 0.99$  when true label  $y_i = 0$  yields massive loss).
- `neg_log_loss`: Inverts the sign ( $-L[Y, P]$ ) so higher values indicate better performance, perfectly matching scikit-learn's maximization conventions.

## Appendix A: Testing on Synthetic Data

### Overcoming the Limitations of Single Historical Paths

- **The Historical Limitation:** A backtest on raw history evaluates only one realized path out of an infinite stochastic process.
- **Synthetic Data Generation:** Generating thousands of statistical twin series to evaluate strategy robustness across unseen market regimes.

### Resampling Approaches

- **Subsampling:** Drawing without replacement.
- **Bootstrapping:** Uniform replacement sampling.
- **Sequential Bootstrap:** Adjusting draw probabilities online by label uniqueness  $u_i^{(t)}$ .

### Monte Carlo Approaches

- **Parametric:** Regime-switching processes (GARCH, Ornstein-Uhlenbeck).
- **Non-Parametric:** Deep Generative Models (Variational Autoencoders - VAEs, GANs).

## Appendix B: Proof of the “False Strategy” Theorem

### Mathematical Foundation of Selection Bias

- **Asymptotic Extreme Value Theory:** The maximum value among independent Gaussian random variables converges asymptotically to a **Gumbel distribution**.
- **The False Strategy Theorem:** Given  $N$  independent trials under  $H_0 : \mu_{SR} = 0$ , the expected maximum Sharpe ratio grows as a function of  $N$ :

$$\mathbb{E} \left[ \max_{n=1, \dots, N} \{\widehat{SR}_n\} \right] \approx \sigma_{SR} \left( (1 - \gamma) Z^{-1} \left[ 1 - \frac{1}{N} \right] + \gamma Z^{-1} \left[ 1 - \frac{1}{Ne} \right] \right)$$

where  $\gamma \approx 0.5772$  is the Euler-Mascheroni constant and  $Z^{-1}[\cdot]$  is the inverse standard normal CDF.

### Takeaway for Quantitative Researchers

- High in-sample Sharpe ratios are mathematically guaranteed if enough trials  $N$  are conducted on random noise.
- Reporting a backtest without reporting  $N$  is equivalent to presenting a statistical illusion.

## Summary: Key Takeaways on Tuning & Synthetic Data

### Core Principles for Financial ML Practitioners

- 1 **Always Purge Hyper-Parameter Tuning:** Never run `GridSearchCV` or `RandomizedSearchCV` without passing a `PurgedKFold` generator with embargoing.
- 2 **Prefer Log-Loss over Accuracy:** Use `neg_log_loss` for tuning to heavily penalize confident false signals and align model probability calibration with PnL risk.
- 3 **Use Log-Uniform Distributions:** When tuning continuous scale parameters  $(C, \gamma, \alpha)$ , sample from log-uniform distributions to span order-of-magnitude ranges efficiently.
- 4 **Test on Synthetic Datasets:** Complement historical backtests with Monte Carlo and resampled synthetic paths to ensure strategies do not rely on historical flukes.
- 5 **Respect the False Strategy Theorem:** Always record total trial count  $N$  and compute Deflated Sharpe Ratio (DSR) to filter out selection bias.

## Q&A

Thank You!

Questions & Discussion

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